United States Patent	12416932
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Masteller; Jeffrey Campbell et al.

Pressure regulators with adjustable boost bodies

Abstract

Pressure regulators with adjustable boost bodies are described herein. An example pressure regulator includes a valve body; a seat in the fluid passageway; an actuator casing coupled to the valve body; and a plug assembly including: a stem; a plug coupled to the stem; a stem guide disposed in the channel of the actuator casing; a boost body on the second side of the stem guide; and a tube extending through the boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the second end disposed in a downstream location relative to the seat, wherein the boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.

Inventors: **Masteller; Jeffrey Campbell (McKinney, TX), Forrest; Keltan Travis (Richardson, TX), Masias; Justin Lane (McKinney, TX), Da Costa; Andrew (Plano, TX), Longoria; Ruben (Little Elm, TX), Ferguson; Alexander John (Denton, TX)**

Applicant: **Emerson Process Management Regulator Technologies, Inc.** (McKinney, TX)

Family ID: **1000008820314**

Assignee: **Emerson Process Management Regulator Technologies, Inc. (McKinney, TX)**

Appl. No.: **18/584510**

Filed: **February 22, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240361787 A1	Oct. 31, 2024

Related U.S. Application Data

continuation-in-part parent-doc US 18475985 20230927 PENDING child-doc US 18584510
us-provisional-application US 63498970 20230428

Publication Classification

Int. Cl.: **G05D16/00** (20060101); **F16K31/365** (20060101)

U.S. Cl.:

CPC **G05D16/024** (20190101); **F16K31/365** (20130101);

Field of Classification Search

CPC: G05D (16/0688); G05D (16/02); G05D (16/0694); G05D (16/063); G05D (16/0683); G05D (16/103); G05D (16/024); Y10T (137/7819); Y10T (137/7831); Y10T (137/7801); Y10T (137/7809); Y10T (137/7812); Y10T (137/7818); F16K (31/165); F16K (31/365)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3086548	12/1962	Galiger	N/A	N/A
4491149	12/1984	Trinkwalder	N/A	N/A
4715232	12/1986	Buettner	N/A	N/A
4754778	12/1987	Duffy	N/A	N/A
5313980	12/1993	Carlson	N/A	N/A
5622207	12/1996	Frank	137/561A	E03B 7/12
8240327	12/2011	Hawkins	137/489.5	F16K 1/00
8281803	12/2011	Mevius	137/505.46	G05D 16/0694
9354639	12/2015	Fan	N/A	N/A
9678516	12/2016	Fan et al.	N/A	N/A
10001785	12/2017	Sanders	N/A	G05D 7/012
10168717	12/2018	Lin	N/A	G05D 16/0688
10247320	12/2018	Zhou	N/A	G05D 16/0683
2009/0261281	12/2008	Mevius	251/118	G05D 16/0694
2010/0071786	12/2009	Hawkins	137/485	F16K 17/32
2011/0174381	12/2010	Foust	N/A	N/A
2013/0228240	12/2012	Noceti	N/A	N/A
2013/0255791	12/2012	Nashery	N/A	N/A
2014/0083530	12/2013	Fan	N/A	N/A
2014/0090727	12/2013	Nguyen et al.	N/A	N/A
2015/0234395	12/2014	Fan et al.	N/A	N/A
2015/0345654	12/2014	Carmen et al.	N/A	N/A
2016/0069469	12/2015	Sanders	73/714	G05D 16/02

2018/0046204	12/2017	Lin	N/A	G05D 16/0688
2020/0133317	12/2019	Wang	N/A	G05D 16/0691

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
3497533	12/2020	EP	N/A
2466800	12/1980	FR	N/A

OTHER PUBLICATIONS

International Searching Authority, “International Search Report,” issued in connection with International Patent Application No. PCT/US2022/035144 mailed on Oct. 11, 2022, 2 pages. cited by applicant

International Searching Authority, “Written Opinion of the International Searching Authority,” issued in connection with International Patent Application No. PCT/US2022/035144, dated Oct. 11, 2022, 6 Pages. cited by applicant

United States Patent and Trademark Office, “Non-Final Office Action”, issued in connection with U.S. Appl. No. 17/459,142, mailed on Feb. 13, 2023, 13 pages. cited by applicant

United States Patent and Trademark Office, “Notice of Allowance,” issued in connection with U.S. Appl. No. 17/459,142, mailed on Mar. 22, 2023, 8 pages. cited by applicant

United States Patent and Trademark Office, “Non-Final Office Action,” issued in connection with U.S. Appl. No. 18/355,845, dated Feb. 9, 2024, 14 pages. cited by applicant

International Searching Authority, “International Preliminary Report on Patentability,” issued in connection with International Application No. PCT/US2022/035144, dated Feb. 27, 2024, 7 pages. cited by applicant

United States Patent and Trademark Office, “Notice of Allowance,” issued in connection with U.S. Appl. No. 18/355,845, dated Apr. 10, 2024, 8 pages. cited by applicant

European Patent Office, “Extended European Search Report,” issued in connection with European Patent Application No. 24172263.6-1201, mailed on Sep. 25, 2024, 8 pages. cited by applicant

European Patent Office, “Communication pursuant to Article 94(3) EPC,” issued in connection with European Patent Application No. 22 748 124.9, dated Dec. 19, 2024, 5 pages. cited by applicant

United States Patent and Trademark Office, “Non-Final Office Action,” issued in connection with U.S. Appl. No. 18/475,985, dated Jan. 27, 2025, 14 pages. cited by applicant

Canadian Intellectual Property Office, “Examiner requisition,” issued in connection with Canadian Patent Application No. 3,229,947, dated Mar. 11, 2025, 4 pages. cited by applicant

European Patent Office, “Communication pursuant to Article 94(3) EPC,” issued in connection with European Patent Application No. 22 748 124.9, dated Apr. 7, 2025, 6 pages. cited by applicant

United States Patent and Trademark Office, “Notice of Allowance and Fee(s) Due,” issued in connection with U.S. Appl. No. 18/475,985, dated Mar. 27, 2025, 7 pages. cited by applicant

---

Primary Examiner: Cahill; Jessica

Attorney, Agent or Firm: Hanley, Flight & Zimmerman, LLC

Background/Summary

RELATED APPLICATIONS (1) This patent arises from a continuation-in-part of U.S. patent application Ser. No. 18/475,985, which was filed on Sep. 27, 2023, and which claims the benefit of U.S. Provisional Patent Application No. 63/498,970, filed on Apr. 28, 2023. U.S. patent application Ser. No. 18/475,985 and U.S. Provisional Patent Application No. 63/498,970 are hereby incorporated herein by reference in their entireties. Priority to U.S. patent application Ser. No. 18/475,985 and U.S. Provisional Patent Application No. 63/498,970 is hereby claimed.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to process control devices and, more particularly, to pressure regulators with adjustable boost bodies.

BACKGROUND

(2) Pressure regulators, sometimes referred to as fluid regulators, are commonly distributed throughout process control systems to control flow rates and/or pressures of various fluids (e.g., liquids, gases, etc.). Pressure regulators can be used to regulate the pressure of a fluid to a substantially constant value. For example, a pressure regulator typically has an inlet that receives a supply of process fluid at a relatively high pressure and reduces the pressure to a lower and/or substantially constant pressure at an outlet.

SUMMARY

(3) A pressure regulator disclosed herein includes a valve body defining a fluid passageway between an inlet and an outlet, a seat in the fluid passageway, and an actuator casing coupled to the valve body. The actuator casing defines a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body. The pressure regulator also includes a plug assembly including a stem, a plug coupled to the stem, and a stem guide disposed in the channel of the actuator casing. The stem is moveable in the stem guide to move the plug relative to the seat. The stem guide has a first side facing the pressure sense chamber and a second side facing the fluid passageway. The plug assembly also includes a boost body on the second side of the stem guide and a tube extending through the boost body. The tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body. The tube has a first end and a second end. The second end disposed in a downstream location relative to the seat such that the pressure sense chamber is in fluid communication with the downstream location. The boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.

(4) A pressure regulator disclosed herein includes a valve body defining a fluid passageway between an inlet and an outlet, a seat in the fluid passageway, and an actuator casing coupled to the valve body. The actuator casing defines a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body. The pressure regulator also includes a plug assembly including a stem, a plug coupled to the stem, and a stem guide disposed in the channel of the actuator casing. The stem is moveable in the stem guide to move the plug relative to the seat. The stem guide has a first side facing the pressure sense chamber and a second side facing the fluid passageway. The stem guide has a flange with a notch. The plug assembly also includes a means for orienting a boost body with a tube from a first orientation to a second orientation relative to the valve body.

(5) A plug assembly for a pressure regulator disclosed herein includes a boost body on a side of a stem guide and a tube extending through a boost body. The tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body. The tube having a first end and a second end. The second end to be disposed in a downstream location relative to a seat in a fluid passageway such that a pressure sense chamber is in fluid communication with the downstream location. The boost body is moveable away from the side of the stem guide and rotatable relative to the stem guide to position the tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a cross-sectional view of an example pressure regulator having an example plug assembly with an example adjustable boost body constructed in accordance with the teachings of this disclosure.

(2) FIG. **2**A-**2**D show the example pressure regulator of FIG. **1** with an example valve body in different orientations.

(3) FIG. **3** is a perspective view of the example pressure regulator of FIG. **1** with the example valve body removed.

(4) FIG. **4** is a side view of the example pressure regulator of FIG. **3** with the example boost body and an example tube removed.

(5) FIGS. **5**A-**5**C are perspective views of the example plug assembly of FIG. **1** showing an example sequence of adjusting a position of the example tube of the example plug assembly.

(6) FIG. **6** is a perspective view of the example boost body and the example tube of the example plug assembly of FIG. **1**.

(7) FIG. **7** is a cross-sectional view taken along line A-A of FIG. **6**.

(8) FIG. **8** is another example pressure regulator with an example pressure regulator having a different stem guide and boost body configuration.

(9) FIG. **9** is a perspective view of the example boost body and example tube of the example plug assembly of FIG. **8**.

(10) FIGS. **10**A-**10**B depict an example boost body, an example tube, and an example locating pin.

(11) FIGS. **11-12** depict example unitary tube and boost body structures.

(12) FIG. **13** is an example unitary tube, boost body, and stem guide.

(13) FIG. **14** is an example key for use with the example of FIG. **13**.

(14) FIG. **15** is an example implementation of the example unitary tube, boost body, and stem guide of FIG. **13** and the example key of FIG. **14**.

(15) FIGS. **16**A-**21**B are additional example unitary tube and boost body components.

(16) The figures are not to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts.

DETAILED DESCRIPTION

(17) In general, pressure regulators (sometimes referred to as fluid regulators) modulate the flow of fluid in accordance with a sensed downstream pressure to maintain process system pressures within acceptable and/or constant pressure limits. Pressure regulators typically include a plug assembly having a stem guide, a stem slidably disposed in the stem guide, and a plug coupled to the stem. The stem guide is disposed in a channel formed in an actuator casing of the pressure regulator. The pressure regulator includes a valve body, which is coupled to the actuator casing, and defines a fluid passageway between an inlet and an outlet. The stem is moveable to move the plug relative to a seat in the fluid passageway to control the flow of fluid through the fluid passageway between the inlet and the outlet. Pressure regulators also include a diaphragm coupled to the stem. The diaphragm separates a pressure sense chamber from a control chamber. The pressure sense chamber is fluidly coupled to and receives fluid from a downstream point. The stem is disposed between the pressure sense chamber and the fluid passageway. The diaphragm moves in response to a difference in force between a force exerted by the pressure of the fluid in the pressure sense chamber (from the sensed downstream point) and the control chamber (e.g., set via a spring). Movement of the diaphragm causes the plug to move away from or toward the seat to allow or restrict the flow of

fluid through the fluid passageway and, thus, between the inlet and the outlet of the pressure regulator.
(18) In some examples, the pressure sense chamber is fluidly coupled by a boost body to a location in the fluid passageway that is immediately downstream of the seat in the valve body of the pressure regulator. This is often referred to as an internal sense pressure regulator. However, the pressure in this location is often relatively high, even when the pressure regulator is closed. This can cause a delay in opening the pressure regulator. As such, the pressure regulator may not quickly respond to changes in pressure.
(19) Disclosed herein are example plug assemblies that include a tube (e.g., a sense tube) that extends through the boost body and is aligned with a notch in a flange of the stem guide. The tube forms a passageway that places the pressure sense chamber in fluid communication with a point that is further downstream in the fluid passageway. Therefore, the pressure sense chamber senses pressure that is further downstream of the seat than known boost bodies. This location further downstream has a lower pressure and higher velocity, which enables the pressure regulator to open more quickly to compensate for mechanical droop of the regulator and allow for the measured pressure to increase.
(20) In some examples, the tube includes a curved portion that forms a 90° turn or bend, which enables the tube to extend relatively far down the center of the fluid passageway downstream of the seat. In some examples, the boost body is adjustable to change the direction that the tube curves relative to the actuator casing. This enables the actuator casing and the valve body to be coupled at different orientations while still enabling the tube to extend down the middle of the downstream portion of the valve body. In some examples, the stem guide has multiple notches, and the boost body can be rotated to align the tube with the different notches depending on the desired orientation of the tube. Therefore, a relatively long tube can be used, while still enabling the actuator casing and the valve body to be coupled at different rotational orientations. This allows the actuator casing and valve body to be coupled at different orientations, which is desirable when installing the pressure regulator between pipes.
(21) FIG. **1** is a cross-sectional view of an example pressure regulator **100** constructed in accordance with the teachings of this disclosure. The example pressure regulator **100** can be used to regulate and/or otherwise control the flow and/or pressure of a process fluid. The process fluid can be any type of fluid, such as natural gas, oil, water, etc. For example, FIG. **1** shows the example pressure regulator **100** installed between an upstream pipe **102** and a downstream pipe **104**. The upstream pipe **102** supplies process fluid from an upstream source (e.g., a distribution facility), and the downstream pipe **104** directs the process fluid to a downstream location (e.g., a customer). The pressure regulator **100** can be used to regulate and/or control the pressure of the fluid between the upstream pipe **102** and the downstream pipe **104**. For example, the pressure regulator **100** can be used to reduce the pressure from a first pressure P**1** in the upstream pipe **102** to a second pressure P**2** in the downstream pipe **104**. The downstream pressure P**2** may be based on the downstream location capacity and/or demands.
(22) In the illustrated example of FIG. **1**, the pressure regulator **100** includes a device body **106** (e.g., a casing, a housing, etc.). In this example, the device body **106** includes a valve body **108** (sometimes referred to as a regulator body or housing) and an actuator casing **110** that are coupled together (e.g., via one or more bolts). However, in other examples, the device body **106** can include more or fewer bodies or housings. In the illustrated example, the valve body **108** defines a fluid passageway **112** between an inlet **114** at a first end **116** of the valve body **108** and an outlet **118** at a second end **120** of the valve body **108**. The upstream pipe **102** is coupled to the first end **116** of the valve body **108** at the inlet **114**, and the downstream pipe **104** is coupled to the second end **120** of the valve body **108** at the outlet **118**. In some examples, the downstream pipe **104** is threaded into the valve body **108** at the outlet **118**. In the illustrated example, the actuator casing **110** defines a pressure sense chamber **122**. The actuator casing **110** also defines a channel **124** (e.g., an opening, a

passageway, etc.) between the fluid passageway **112** and the pressure sense chamber **122**.
(23) In the illustrated example, the pressure regulator **100** includes a seat **126** (e.g., a seal) in the fluid passageway **112**. The pressure regulator **100** also includes an example plug assembly **128**. The plug assembly **128** controls the flow of fluid through the seat **126** and, thus, between the inlet **114** and the outlet **118**. The seat **126** divides the fluid passageway **112** into an upstream portion **130** (upstream of the seat **126**) and a downstream portion **132** (downstream of the seat **126**).
(24) In the illustrated example, the plug assembly **128** includes a plug **134** (sometimes referred to as a disc or flow control member), a stem **136**, and a stem guide **138**. The plug **134** is coupled to the stem **136**. In the illustrated example, the stem **136** extends through the stem guide **138**. The stem **136** is moveable (e.g., slidable) in the stem guide **138** to move the plug **134** toward or away from the seat **126**, thereby controlling the flow of fluid through the seat **126**. In particular, the plug **134** can be moved between a closed position, which is shown in FIG. **1**, and an open position. In the closed position, the plug **134** is sealingly engaged with the seat **126**, which prevents fluid flow through the seat **126** from the inlet **114** to the outlet **118**. In the open position, the plug **134** is moved to the left in FIG. **1** such that the plug **134** is spaced from the seat **126**, which allows fluid to flow through the seat **126** from the inlet **114** to the outlet **118**. Therefore, the seat **126** and the plug **134** form a valve. The pressure regulator **100** is configured to move the plug **134** between the opened and closed positions to regulate the pressure of the fluid downstream.
(25) In the illustrated example, the stem guide **138** is disposed in the channel **124** between the fluid passageway **112** and the pressure sense chamber **122**. The stem guide **138** is sealed relative to the device body **106** to fluidly isolate the fluid in the fluid passageway **112** and fluid in the pressure sense chamber **122** (except for designated passages for pressure sense, disclosed in further detail herein). In some examples, the stem guide **138** is sized to form an interference fit (sometimes referred to as friction fit or press fit) with an inner surface **139** of the channel **124**. Additionally or alternatively, one or more seals can be used. The stem guide **138** has a first side **140** and a second side **142** opposite the first side **140**. The first side **140** faces the pressure sense chamber **122** and is exposed to the pressure sense chamber **122**. The second side **142** faces the fluid passageway **112**.
(26) In this example, the example pressure regulator **100** balances the stem **136** to reduce or eliminate the influence of the pressure differential on the stem **136** and the plug **134**. In the illustrated example, the stem **138** has a stem body **144** and a flange **146** extending radially from the stem body **144**. The stem guide **138** and the flange **146** form a balance chamber **148**. The flange **146** is sealed to an inner surface of the stem guide **138** via a seal **150**. As such, the flange **146** fluidly separates the balance chamber **148** and the downstream portion **132** of the fluid passageway **112**.
(27) In the illustrated example, a balance passage **152** (which may also be referred to as a balance port) is defined through the plug **134** and the stem **136** that extends to the balance chamber **148**. In particular, a portion of the balance passage **152** extends longitudinally through the stem **136**, and another portion of the balance passage **152** extends in a transverse direction (e.g., into or out of the page in FIG. **1**) to the outer surface of the stem **136**. Upstream or inlet fluid pressure is delivered to the balance chamber **148** through the balance passage **152**. Thus, upstream or inlet fluid pressure acts on the stem **136** in the balance chamber **148** to counter the upstream or inlet fluid pressure acting on the plug **134**. Therefore, in this example, the balance passage **152** forms means for fluidly coupling the upstream portion **130** of the fluid passageway **112** and the balance chamber **148**. In other examples, the balance passage **152** could be formed by one or more openings or channels through one or more other structures.
(28) As shown in FIG. **1**, the pressure regulator **100** includes a control spring **154**. The control spring **154** is used to control or set the pressure at which the pressure regulator **100** opens and closes. The control spring **154** is disposed in a control chamber **156** in the actuator casing **110**. In some examples, the control chamber **156** is open to the atmosphere via a vent opening **158**. In other examples, a fluid line can be coupled to the vent opening **158** to place the control chamber **156** at a

different pressure than atmosphere. In the illustrated example, the pressure regulator **100** includes a diaphragm **160** that separates the control chamber **156** and the pressure sense chamber **122**. The pressure regulator **100** includes a diaphragm plate **162** that is coupled to the diaphragm **160** via a diaphragm cap screw **164**. The control spring **154** is engaged with the diaphragm plate **162**, which biases the diaphragm plate **162** and the diaphragm **160** toward the pressure sense chamber **122** (downward in FIG. **1**). The diaphragm plate **162** is coupled to a pusher post **166** (e.g., via the diaphragm cap screw **164**). The pusher post **166** includes a drive pin **168** that cooperates with a lever **170**. The lever **170** is pivotably coupled to the actuator casing **110**. The lever **170** is coupled to a stem pin **172** that is coupled to the stem **136**. As the pusher post **166** moves up and down, the pusher post **166** rotates the lever **170**, which moves the stem **136** linearly in the stem guide **138** (left and right in FIG. **1**).

(29) The pressure regulator **100** receives fluid at the inlet **114** at the first pressure P**1**. The pressure regulator **100** is configured to stop or reduce the flow of fluid to the outlet **118** based on pressure of fluid at a downstream point, referred to as the second pressure P**2**. If the pressure P**2** meets or exceeds a certain pressure (which may be referred to herein as a set or trigger pressure), the pressure regulator **100** closes the fluid passageway **112**, thereby regulating the pressure of the fluid at the downstream point. In the illustrated example, the pressure sense chamber **122** receives fluid from downstream of the seat **126** via a boost body and tube, as disclosed in further detail herein. Therefore, the pressure sense chamber **122** is at the second pressure P**2**. When the pressure P**2** in the pressure sense chamber **122** reaches the set pressure, the pressure P**2** overcomes the force of the control spring **154** and forces the diaphragm **160** upward in FIG. **1**. As the diaphragm **160** and the diaphragm plate **162** move upward, the pusher post **166** rotates the lever **170** clockwise in FIG. **1**. The lever **170** pushes the stem **136** to the right in FIG. **1**, until the plug **134** engages the seat **126** to close off the fluid passageway **112**, as shown in the position in FIG. **1**. This reduces the pressure of the fluid at the downstream point. When the pressure P**2** of the fluid in the pressure sense chamber **122** drops below the set pressure, the spring force overcomes the pressure and moves the diaphragm **160** and the diaphragm plate **162** downward. The pusher post **166** moves downward and rotates the lever **170** in the opposite direction. In some examples, the pressure regulator **100** includes a return spring **174** to bias the stem **136** back to the left in FIG. **1**, such that the plug **134** moves away from the seat **126**. Therefore, the set pressure is based on the spring force from the control spring **154**. In some examples, the spring force of the control spring **154** can be increased or decreased to change the set pressure to a desired pressure level. For example, an adjustment screw **176** can be adjusted to increase or decrease the compression of the control spring **154** and, thus, change the force the control spring **154** exerts on the diaphragm **160**.

(30) As disclosed above, the pressure sense chamber **122** is fluidly coupled and/or otherwise senses pressure at a location that is downstream of the seat **126**. As shown in FIG. **1**, the plug assembly **128** includes a boost body **178** and a tube **180** (which may also be referred to as a pitot tube or sense tube). The boost body **178** is disposed in the channel **124** on the second side **142** of the stem guide **138**. The boost body **178** is sealed against the second side **142** of the stem guide **138** by a seal **179** (e.g., an o-ring). As shown in FIG. **1**, the plug assembly **128** includes a spring **181** to bias the boost body **178** toward the stem guide **138**. In this example, the spring **181** is between the plug **134** and the boost body **178**. In other examples, the spring **181** can be disposed in another location (e.g., between the boost body **178** and a portion of the valve body **108**). In this example, the spring **181** is a conical compression spring. In other examples, the spring **181** can be implemented by another type of spring, such as a constant pitch helical compression spring. The tube **180** is coupled to the boost body **178**. In the illustrated example, a portion of the tube **180** extends through the boost body **178** and into the stem guide **138**. In particular, the stem guide **138** has a first flange **182** and a second flange **184**. The second flange **184** is spaced from (in the axial direction) the first flange **182**. The first and second flanges **182**, **184** can also be referred to as discs. The first and second flanges **182**, **184** are engaged (e.g., in contact) with the inner surface **139** of the channel

124. The first flange **182** has a set of notches **186***a*-**186***d* (the first and second notches **186***a* and **186***c* are shown in FIG. **1**) that extend into the peripheral edge of the first flange **182**. The second flange **184** has a notch **188** to enable fluid flow across the second flange **184**. The tube **180** has a first end **190** and a second end **192** opposite the first end **190**. As shown in FIG. **1**, the tube **180** extends through the boost body **178** and is aligned with the first notch **186***a* in the first flange **182** of the stem guide **138**. In particular, in this example, the first end **190** extends into and is disposed in the first notch **186***a*. The second end **192** is disposed in a downstream location of the seat **126**. As such, fluid from the second end **192** of the tube **180** is directed through the tube **180**, through the first notch **186***a*, through the cavity between the stem guide **138** and the inner surface of the channel **124**, and through the notch **188** in the second flange **184** of the stem guide **138** and into the pressure sense chamber **122**. Therefore, the tube **180** fluidly couples the downstream location at the second end **192** of the tube **180** with the pressure sense chamber **122**. As such, the pressure sense chamber **122** is at the same pressure as the fluid at the downstream location where the second end **192** of the tube **180** is located.
(31) In this example, the tube **180** is curved and includes a 90° turn or bend (e.g., a curve or bend). This enables the tube **180** to extend further down the fluid passageway **112** away from the seat **126**. In some examples, this is advantageous because the pressure further downstream is sensed as a lower pressure. This enables the pressure regulator **100** to open faster or earlier by reducing feedback delay to compensate for mechanical droop of the pressure regulator **100**. In known regulators, the pressure sense chamber **122** senses the pressure of the fluid immediately adjacent to the seat **126**. However, the pressure next to the seat **126** is higher than the pressure downstream as the pressure regulator **100** opens and allows more flow. This higher pressure maintains the pressure regulator **100** closed longer. By extending the tube **180** further downstream, as shown in FIG. **1**, the pressure sense chamber **122** can sense a lower pressure than immediately adjacent to the seat **126**. This enables the pressure regulator **100** to open more quickly by accounting or compensating for mechanical droop.
(32) In some examples, the second end **192** of the tube **180** is located at or near the second end **120** of the valve body **108**. In the illustrated example, the second end **192** of the tube **180** extends beyond or past the second end **120** of the valve body **108**. As such, a portion of the tube **180** extends into the downstream pipe **104**. In some examples, it is advantageous to sense the pressure just outside of the valve body **108**, because this pressure registers as a lower pressure to achieve the benefits discussed above. However, in other examples, the tube **180** may be longer or shorter such that the tube **180** extends further into the downstream pipe **104**, or the second end **192** may be in the downstream portion **132** of the fluid passageway **112**.
(33) In some examples, the valve body **108** and the actuator casing **110** can be coupled to each other in different orientations. For example, FIGS. **2**A-**2**D show four different orientations of the valve body **108** relative to the actuator casing **110**. The orientation may be chosen by an installer for a specific installation. In some examples, the desired orientation may be based on the existing pipe setup and the desired orientation of the valve body **108** and/or the actuator casing **110** relative to each other. For example, it may be desirable to orient the actuator casing **110** vertically, while the valve body **108** can be orientated to match the direction of the upstream and downstream pipes. In another example, the pressure regulator **100** may be connected to a pre-existing pipe, and it may be desirable to orient the actuator casing **110** such that the vent opening **158** is facing vertically downward, which helps prevent rain or other precipitation from entering the vent opening **158**. In this example, the valve body **108** and the actuator casing **110** can be coupled in four different orientations (e.g., at 90° increments), which may be based on a bolt pattern between the valve body **108** and the actuator casing **110**.
(34) As disclosed above, the tube **180** extends in a certain direction away from the boost body **178** (e.g., upward in FIG. **1**). For example, FIG. **3** is a perspective view of the pressure regulator **100** with the valve body **108** removed. As shown in FIG. **3**, the actuator casing **110** has four flange bolt

openings **300** (one of which is referenced in FIG. **3**). The valve body **108** (FIG. **1**) can be coupled to the actuator casing **110** in one of four orientations (shown in FIGS. **2**A-**2**D) based on the bolt pattern. As shown, the tube **180** is curved upward in the orientation in FIG. **3**. Therefore, to rotate the valve body **108** to another orientation (e.g., horizontal), the tube **180** is reoriented so the tube **180** can remain centered in the downstream portion **132** (FIG. **1**) of the fluid passageway **112** (FIG. **1**).

(35) FIG. **4** is a side view of the pressure regulator **100** of FIG. **3** with the boost body **178** and the tube **180** removed. As disclosed above, the first flange **182** of the stem guide **138** includes the four notches **186*a*-186*d***. This enables the tube **180** to be inserted into other ones of the notches **186*a*-186*d*** depending on the desired orientation of the tube **180**. Therefore, the tube **180** can be moved to other positions and/or orientations relative to the stem guide **138**. In the illustrated example, the notches **186*a*-186*d*** extend into a peripheral edge **400** of the first flange **182** of the stem guide **138**. However, in other examples, the notches **186*a*-186*d*** may instead be formed by openings (e.g., through-holes, channels) that extend through the first flange **182** and do not extend to the peripheral edge **400**.

(36) FIGS. **5**A-**5**C are perspective views of the plug assembly **128** showing an example sequence of adjusting the orientation of the tube **180**. In FIG. **5**A, the boost body **178** is engaged or sealed against the stem guide **138** and the first end **190** of the tube **180** is disposed in the first notch **186*a***. In this position, the tube **180** extends outward and vertically upward. Also shown and labeled in FIG. **5**A are the notch **188** in the second flange **184** of the stem guide **138**, the return spring **174**, the stem **136**, the stem pin **172**, and the plug **134**.

(37) To change the orientation of the boost body **178** and the tube **180**, the boost body **178** can be moved away from the stem guide **138**, against the bias of the spring **181**. For example, as shown in FIG. **5**B, the boost body **178** can be pulled linearly away (in the direction of the arrow) from the stem guide **138** such that the first end **190** of the tube **180** is moved out of the first notch **186*a***. In some examples, an installer (e.g., a person) can manually pull the boost body **178** away from the stem guide **138**. Then, as shown in FIG. **5**C, the boost body **178** with the tube **180** can be rotated (e.g., clockwise or counter-clockwise) to align the tube **180** with another notch **186*b*-186*d***. When the tube **180** is in the desired orientation and aligned with one of the notches **186*a*-186*d***, the boost body **178** can be released, and the spring **181** biases the boost body **178** toward the stem guide **138** such that the first end **190** of the tube **180** is inserted into the corresponding notch. The spring **181** ensures the parts stay together as intended by the installer during installation. Therefore, the boost body **178** is rotatable with respect to the stem guide **138** to position the tube **180** in at least one other orientation relative to the actuator casing **110** (FIG. **1**). In this example, the boost body **178** and the tube **180** can be rotated to one of four positions or orientations to align the first end **190** of the tube **180** with one of the four notches **186*a*-186*d***. In this example, a portion of the tube **180** extends into the corresponding notch **186*a*-186*d***. Therefore, the tube **180** acts as a locating or indicating pin to align the tube **180** in one of the four positions. Once the tube **180** is in the desired orientation, the valve body **108** (FIG. **1**) can be installed on the actuator casing **110** (FIG. **1**) (or vice versa). As such, the orientation of the tube **180** can be changed while the plug assembly **128** remains in the channel **124** of the actuator casing **110**.

(38) In this example, the four notches **186*a*-186*d*** are radially spaced apart from each other around the first flange **182**. In particular, in this example, the notches **186*a*-186*d*** are spaced 90° apart from each other. As such, the tube **180** can be rotated and positioned at one of four different rotational positions that are 90° apart. These four rotational positions correspond to the rotational positions in which the valve body **108** and the actuator casing **110** can be coupled, as shown in FIGS. **2**A-**2**D. However, in other examples, the stem guide **138** can include more or fewer notches and/or the notches can be spaced apart at other increments (e.g., **6** notches spaced apart at 60° increments).

(39) In some examples, the plug assembly **128** is sized and shaped to fit in existing pressure regulators. Therefore, old plug assemblies can be removed and replaced with the example plug

assembly **128** with the tube **180**.
(40) FIG. **6** is a perspective view of the boost body **178** and the tube **180**. The boost body **178** has a first side **600**, a second side **602**, and a peripheral edge **604** between the first side **600** and the second side **602**. The tube **180** extends through a channel **606** formed between the first and second sides **600**, **602** of the boost body **178**. In some examples, the tube **180** and the channel **606** may be dimensioned such that the tube **180** fits snuggly (e.g., forms an interference fit (sometimes referred to as a friction fit or press fit)) within the channel **606** to prevent fluid leaking through the channel **606**. For example, the diameter of the tube **180** may be larger than the diameter of the channel **606** prior to insertion into the channel **606**. In some examples, the boost body **178** is constructed of a compliant material, such as rubber, and the tube **180** is constructed of metal, such as stainless steel.
(41) In this example, the tube **180** is coupled to the boost body **178** by a retaining pin. For example, FIG. **6** shows a bore **608** extending into the peripheral edge **604** of the boost body **178**. The bore **608** partially intersects the channel **606**. For example, FIG. **7** is a cross-sectional view of the boost body **178** taken along line A-A of FIG. **6**. As shown in the callout in FIG. **7**, a pin **700** is inserted into the bore **608**. When the pin **700** is inserted into the bore **608**, the pin **700** partially crushes or flattens a portion of the tube **180**, as shown in the callout. This locks and/or otherwise prevents the tube **180** from being pulled out of the boost body **178**. To remove the tube **180** from the boost body **178**, the pin **700** is removed first, and then the tube **180** can be slid outward from the channel **606**. In other examples, the tube **180** can be coupled to the boost body **178** via other techniques (e.g., a clamp, a threaded fastener). Further, in some examples, the boost body **178** and the tube **180** can be constructed (e.g., machined) as a single unitary part or component (e.g., a monolithic structure). As used herein, a unitary structure is defined as a structure that is monolithic.
(42) In the example of FIG. **7**, the tube **180** is held in place by the pin **700** in the boost body **178**. While friction between the pin **700** and the boost body **178** tends to resist movement of the pin **700** relative to the boost body **178**, the tube **180** is not prevented from, for example, rotating a small amount relative to the pin **700** and, thus, the boot body **178**. Additionally, when the tube **180** is compressed by the pin **700** in the assembly process, the pressure from the pin **700** can cause the tube **180** to move relative to the boost body **178**. Still further, when the tube **180** and the boost body **178** are separately manufactured parts, the assembly requires the machining of the boost body **178** (e.g., the plate), the bending of the tube **180**, and the installation of the tube **180** and the pin **700** into the boost body **178**. Constructing the example tube **180** and the example boost body **178** as a unitary piece eliminates the need for the pin **700** of FIG. **7**, the related assembly operations and, advantageously, the possibility of movement (e.g., rotation) of the pin **700** relative to the boost body **178** such as mentioned above.
(43) In some examples described herein, the tube **180** and boost body **178** assembly may be formed using additive manufacturing (i.e., 3-D printing). When the boost body **178** (e.g., the plate) and the tube **180** are printed as a single, unitary piece, the assembly requires less machining because the channel **606** and the bore **608** are eliminated.
(44) FIG. **8** is a cross-sectional view of another example pressure regulator **800**. The example pressure regulator **800** is similar to the example pressure regulator **100** disclosed above, and like parts will be given like numbers. However, in this example, the plug assembly **128** includes a stem guide **802** that is different than the stem guide **138** disclosed above. In this example, the stem guide **802** has one channel **804** between a first side **806** and a second side **808** of the stem guide **802**. The tube **180** extends into the channel **804**. As such, fluid from the tube **180** is routed through the stem guide **802** to the pressure sense chamber **122**.
(45) In the example of FIG. **8**, the boost body **178** and the stem guide **802** are coupled by a pin **810**. In particular, the boost body **178** has an opening **812**, and the stem guide **802** has a bore **814**. The pin **810** is disposed in the opening **812** of the boost body **178** and extends into the bore **814**. As such, because both the tube **180** and the pin **810** extend into the stem guide **802**, the boost body **178** and the stem guide **802** are rotationally locked. In other words, the boost body **178** is not rotatable

relative to the stem guide **802** like the example of FIG. **1**. Instead, to change the orientation of the tube **180**, the entire plug assembly **128** can be removed from the channel **124**, rotated, and then reinserted into the channel **124** in the desired orientation. FIG. **9** is a perspective view of the boost body **178** of FIG. **8** showing the opening **812** in the boost body **178**.

(46) FIGS. **10**A-**10**B depict an example boost body **1010**, an example locating pin **1020**, and an example tube **1030**. In the example of FIGS. **10**A-**10**B, the boost body **1010** and the tube **1030** are not unitary. In some examples, the boost body **1010** may be a separate piece or attached to the valve body **108** or the stem guide **138** (shown in FIG. **1**). Further, in the example of FIGS. **10**A-**10**B, the locating pin **1020** is only included for purposes of explaining various alignments of parts and is not used in implementations of the non-unitary assembly of FIGS. **10**A-**10**B.

(47) The position of the locating pin **1020** indicates (e.g., to a person adjusting the position of the tube **1030**) the position or orientation of the tube **1030** relative to the boost body **1010** when a pin (e.g., the pin **700** of FIG. **7**) is used to secure the tube **1030** in the boost body **1010**. The position of the locating pin **1020**, indicated by an axis or line **1040** extending through the center of the locating pin **1020** and the center of an opening **1043** of the boost body **1010**, is intended to indicate the desired position of the tube **1030**. However, as shown in the example of FIGS. **10**A-**10**B, when a pin (e.g., the pin **700** of FIG. **7**) is used to secure the tube **1030** in the boost body **1010**, the tube **1030** may become skewed relative to the locating pin **1020**. In the example of FIG. **10**A, and as shown in FIG. **7**, a pin (e.g., the pin **700** of FIG. **7**) is used to secure the tube **1030** to the boost body **1010**. A line **1042** represents the orientation of the tube **1030** after it has been secured by the pin. Specifically, the line **1042** represents the central longitudinal axis of an end **1044** of the tube **1030**. As demonstrated by the difference (e.g., angular displacement or skew) between the line **1040** and the line **1042**, using a pin to secure the tube **1030** to the boost body **1010** can cause the tube **1030** to move or skew. Because the tube **1030** is not properly aligned with the locating pin **1020** (i.e., the line **1042** is not coincident with the line **1040**), the position of the locating pin **1020** no longer accurately indicates the downstream location or orientation of the tube **1030**.

(48) As shown in FIG. **10**B, the pin (e.g., the pin **700** of FIG. **7**) causes another angular offset or skew of the tube **1030**. This angular offset is demonstrated, in the orientation of FIG. **10**B, by the skew between the line **1040** and the line **1042**. Further, in this view, the locating pin **1020** has a central longitudinal axis **1022** that intersects the line **1040** at a 90 degree angle (i.e., is perpendicular to the line **1040**). As demonstrated in FIG. **10**B, the line **1042** does not intersect at a 90 degree angle with the axis **1022**. Therefore, because the pin skews the position of the tube **1030** with respect to the boost body **1010**, the locating pin **1020** does not accurately represent the downstream position of the tube **1030**.

(49) FIG. **11** is an example implementation of a unitary structure in accordance with the teachings herein. In the example of FIG. **11**, a boost body **1110** and a tube **1120** form the unitary structure. In FIG. **11**, the tube **1120** has a first longitudinal axis **1140** that is perpendicular to a surface **1142** of the boost body **1110**. In other words, the tube **1120** is perpendicular to the surface **1142** at the point it intersects the surface **1142**. Further, the first longitudinal axis **1140** of the tube **1120** is parallel to a longitudinal axis **1132** through the center of the boost body **1110**. As a result of the unitary formation of the boost body **1110** and the tube **1120**, a second longitudinal axis **1122** of the tube **1120** intersects the longitudinal axes **1132**, **1140** at a 90 degree angle (i.e., the second longitudinal axis **1122** is perpendicular to the axes **1132** and **1140**).

(50) FIG. **12** is a further example implementation of a unitary structure in accordance with the teachings herein. In the example of FIG. **12**, a boost body **1210** and a tube **1220** form the unitary structure. In the example of FIG. **12**, the tube **1220** is straight and, thus, not curved. The straight tube **1220** is easier to machine. Then during installation, the tube **1220** may be bent or curved based on the installation conditions and/or application. In some examples, the tube **1220** may be comprised of 316L stainless steel, titanium, 17-4 stainless steel, and other metal-printed material. Further, the tube **1220** may be comprised of any material that resists corrosion and is sufficiently

ductile to enable forming (e.g., bending) in the field during installation. In some examples, the tube **1220** may be bent using a tool (e.g., a tubing bender or other suitable tool to bend the tube). Further, in some examples, the tube **1220** may be bent by hand when performing installation.

(51) FIG. **13** is an example of an adjustable tube assembly **1300** with a boost body **1310**, a tube **1320**, and a stem guide **1330**. In this example, the adjustable tube assembly **1300** is a unitary structure. In this example, the stem guide **1330** has four notches **1332**, **1334**, **1336**, **1338**. In some examples, the stem guide **1330** may have one notch, two notches, three notches, or more than four notches. In some examples, the unitary structure **1300** shown in FIG. **13** may be combined with other pieces of the pressure regulator in addition to the boost body **1310** (e.g., valve body, actuator casing, etc.) to further simplify manufacturing the assembly.

(52) A non-unitary assembly of the parts of FIG. **13** relies on friction between the boost body **178** and the valve body **108** to prevent rotation of the non-unitary assembly relative to the lever **170**. Instead, in the example of FIG. **13**, the adjustable tube assembly **1300** may be keyed so that rotation of the assembly **1300** depends on the fit between a key **1400**, described in more detail in connection with FIG. **14**, and the notches **1332**, **1334**, **1336**, **1338** of the stem guide **1330**. In some examples, the notches **1332**, **1334**, **1336**, **1338** may be added to the interior sides of stem guide **1330** and offset relative to the tube **1320**. The notches **1332**, **1334**, **1336**, **1338** on the interior of the stem guide **1330** allow key teeth, projections, or tabs **1410**, **1412**, **1414**, **1416** of a keyed stem **1420** (as shown in FIG. **14**) to be inserted into the adjustable tube assembly **1300**, as shown in FIG. **15**. In some examples, the notches **1332**, **1334**, **1336**, **1338** may be curved for ease of manufacture.

(53) FIG. **14** is detailed view of the example key **1400** to key the orientation of the adjustable tube assembly **1300**. As used herein, keying is the process by which an installer may change the position or orientation of the tube **1320** and the stem guide **1330** (i.e., the attached boost body **1310**) relative to the lever **170** and the valve body **108**. In some examples, the key **1400** may have one key projection, two key projections, three key projections, or more than four key projections. Further, the key projections **1410**, **1412**, **1414**, **1416** may be of varying shapes and sizes to correspond with the adjustable tube assembly **1300**. Further, in this example, the connection **1430** connects the key **1400** to the lever **170**.

(54) FIG. **15** is an example implementation **1500** of the key **1400** keying the adjustable tube assembly **1300**. As described above, keying changes the orientation of the tube **1320** relative to the lever **170** and the valve body **108**. Keying is the process by which the boost body is separated from the key **1400**, rotated to a position (i.e., a predetermined orientation) relative to the key **1400** based on the position of the key units of the tool in relation to the notches **1332**, **1334**, **1336**, **1338** of the boost body (i.e., stem guide), and returned to mate with the key **1400** in the aligned position. As shown in FIG. **15**, a yoke **1430** is inserted through the stem guide **1330** so that the key projections **1410**, **1412**, **1414**, **1416** may mate with corresponding ones of the notches **1332**, **1334**, **1336**, **1338**. In the example of FIG. **15**, the boost body **1310** may be rotated clockwise in 90 degree increments to adjust the location of the tube **1320** relative to the lever **170**. However, in other examples, the stem guide **1330** may be configured to rotate in increments of more or less than 90 degrees and/or may be configured to in unequal increments that are more or less than 90 degrees.

(55) For example, while in this example the notches **1332**, **1334**, **1336**, **1338** are spaced 90 degrees apart to facilitate various 90 degree orientations of the tube **1320**, the notches **1332**, **1334**, **1336**, **1338** may be spaced any distance apart to orient the tube **1320** in other increments (i.e., 45 degree increments, 100 degree increments, etc.). Further, the notches **1332**, **1334**, **1336**, **1338** can be spaced in any increment relative to each other.

(56) FIGS. **16**A-**21**B are various example implementations of a unitary tube and boost body. The examples illustrated herein are non-exhaustive of possible unitary assemblies. Additionally, various implementations may have different tube diameters, tubes with different curves, tubes with different lengths, or assemblies composed of different materials.

(57) FIGS. **16**A-**16**B are perspective views of an example implementation of a unitary tube **1610**

and boost body **1620**. While not shown in the example of FIGS. **16**A-**16**B, the example tube **1610** continues from the bottom side of the boost body **1620**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **1610** connects into the boost body **178** and provides fluid path therethrough. In this example, a portion of the tube **1610** is supported by a first arm **1632** of the boost body (i.e., the boost body **1620**) and a second arm **1634** of the boost body (i.e., the boost body **1620**). Further, in some examples, the first arm **1632** and the second arm **1634** of the boost body (i.e., the boost body **1620**) are unitary with the boost body (i.e., the boost body **1620**) and the tube **1610**. In this example, the arms **1632** and **1634** are gussets or web-like structures that provide support (e.g., strength and rigidity) to the tube **1610**. However, the arms **1632** and **1634** may have other shapes, geometries, etc. to provide the same or a similar function. While there are two arms **1632**, **1634** shown in this example, in other implementations there may be one arm or more than two arms. In this example, the boost body **1620** has two holes **1622**, **1624** that may be utilized to fasten the boost body **1620** to the boost body **178**. In other examples, one hole or more than two holes may be utilized to fasten the boost body **1620** to the boost body **178**. In some examples, the arms **1632**, **1634** may be manufactured using a separate additive manufacturing process than the process to unify the tube **1610** and the boost body **1620**. The separate additive manufacturing process may be employed to print parts on a previously machined piece. In this example, the boost body **1620** has a semicircular shape. However, in other examples, the boost body **1620** may be any other suitable shape (e.g., a circle, an ellipse, an oval, a polygon, etc.). In this example, an end **1612** of the tube **1610** extends away from the boost body **1620**. However, in other examples, the tube **1610** may be formable in a different direction, or, additionally or alternatively, extend in a direction other than that depicted in FIGS. **16**A-**16**B. In this example, the tube **1610** emerges from the midpoint of the semicircular boost body **1620**. In other examples, the tube **1610** may emerge from another location along the boost body **1620**. While FIGS. **16**A-**16**B depict one tube **1610**, more than one tube may be utilized in other implementations.

(58) FIGS. **17**A-**17**B are perspective views of an example implementation of a unitary assembly of an example tube **1710** and an example boost body **1720**. While not shown in the example of FIGS. **17**A-**17**B, the example tube **1710** continues from the bottom side of the boost body **1720**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **1710** connects into the boost body **178**. In this example, a portion of the tube **1710** is curved to form a first portion **1712** and a second portion **1714**, the first portion **1712** is nearest to the boost body (i.e., nearest to the boost body **1720**) and the second portion **1714** is nearest to an end **1713** of the tube **1710**. A fillet or web **1730** connects the first portion **1712** and the second portion **1714**. In this example, the fillet **1730**, the first portion **1712**, and the second portion **1714** form a unitary structure. In this example, the fillet **1730** is a web-like structure that provides structural support (e.g., strength and rigidity) to the tube **1710**. However, the fillet **1730** may have other shapes, geometries, etc. to provide the same or a similar function. Further, in other examples, there may be more than one fillet to further support the tube **1710** (i.e., a fillet positioned between the boost body **1720** and the tube **1710**). In this example, the boost body **1720** has two holes **1722**, **1724** that may be utilized to fasten the boost body **1720** to the boost body **178**. In other examples, one hole or more than two holes may be utilized to fasten the boost body **1720** to the boost body **178**. In some examples, the fillet **1730** may be manufactured using a separate additive manufacturing process than the process to unify the tube **1710** and the boost body **1720**. As mentioned above, the separate additive manufacturing process may be employed to print parts on a previously machined piece. In this example, the boost body **1720** is shaped as a partial ring. However, in other examples, the boost body **1720** may have any other suitable shape (e.g., a straight line, a circle, an ellipse, an oval, a square, etc.). In this example, an end of the tube **1712** points away from the boost body **1720**. However, in other examples, the tube **1710** may be formable in a different direction, or, additionally or alternatively, pointed in a direction other than that depicted in FIGS. **17**A-**17**B. In this example, the tube **1710** emerges from the midpoint of the boost body **1720**. In other examples, the tube **1710** may emerge

from another location along the boost body **1720**. While FIGS. **17**A-**17**B depict one tube **1710**, more than one tube may be utilized in other implementations.

(59) FIGS. **18**A-**18**B are perspective views of an example implementation of a unitary assembly of an example tube **1810** and an example boost body **1820**. While not shown in the example of FIGS. **18**A-**18**B, the example tube **1810** continues from the bottom side of the boost body **1820**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **1810** connects into the boost body **178**. In this example, a portion of the tube **1810** is bent or curved to form a first portion **1812**, a second portion **1814**, and a third portion **1816**, wherein the first portion **1812** is at a first orientation, the second portion **1814** is at a second orientation different than the first orientation, and the third portion **1816** is at a third orientation different than the first and second orientations. In this example, the boost body **1820** has two holes **1822**, **1824** that may be utilized to fasten the boost body **1820** to the boost body **178**. In other examples, one hole or more than two holes may be utilized to fasten the boost body **1820** to the boost body **178**. Additionally, in this example, the cross-section of the tube **1810** has a teardrop shape. The teardrop shaped cross-section is an aerodynamic shape that prevents disruption of the flow of the fluid through the tube **1810**. While this example cross-section is shown in connection with the implementation of FIG. **18**, it may be applied to any tube. Other suitable shapes and cross-sections may be used additionally with or alternatively to the teardrop shaped cross-section. In some examples, the portion of the tube attached to the boost body **1820** up until a first bend in the tube (i.e., the bend nearest to the boost body **1820**) has a teardrop shaped cross-section, and then after the first bend, the tube **1810** has a circular cross-sectional shape. While in this example the tube **1810** is not supported by an arm or web, in other implementations the tube **1810** may be connected to the boost body **1820** by one or more arm(s) or web-like support structures of varying shapes and sizes. In this example, the boost body **1820** is shaped as a partial circle or ring. However, in other examples, the boost body **1820** may have any other suitable cross-sectional shape (e.g., a straight line, a circle, an ellipse, an oval, a square, etc.). In this example, the end of the tube **1818** points away from the boost body **1820**. However, in other examples, the tube **1810** may be formable in a different direction, or, additionally or alternatively, pointed in a direction other than that depicted in FIGS. **18**A-**18**B. In this example, the tube **1810** emerges from the midpoint of the boost body **1820**. In other examples, the tube **1810** may emerge from another location along the boost body **1820**. While FIGS. **18**A-**18**B depict one tube **1810**, more than one tube may be utilized in other implementations.

(60) FIGS. **19**A-**19**B are perspective views of an example implementation of a unitary assembly of an example tube **1910** and an example boost body **1920**. While not shown in the example of FIGS. **19**A-**19**B, the example tube **1910** continues from the bottom side of the boost body **1920**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **1910** connects into the boost body **178**. In this example, the example boost body **1920** contains four holes **1922**, **1924**, **1926**, **1928** that may be utilized to fasten the boost body **1920** to the boost body **178**. In other examples, one hole, two holes, three holes, or more than four holes may be utilized to fasten the boost body **1920** to the boost body **178**. In this example, the tube **1910** is additionally connected to the collar by two arms **1930**, **1932**. In some examples, the arms **1930**, **1932** may be manufactured using a separate additive manufacturing process than the process to unify the tube **1910** and the boost body **1920**. The separate additive manufacturing process may be employed to print parts on a previously machined piece. While there are two arms **1930**, **1932** shown in this example, in other implementations there may be one arm or more than two arms. In this example, the arms **1930** and **1932** are web-like structures that provide structural support (e.g., strength and rigidity) to the tube **1910**. However, the arms **1930** and **1932** may have other shapes, geometries, etc. to provide the same or a similar function. In this example, the boost body **1920** has a circular cross-sectional shape. However, in other examples, the boost body **1920** may be any other suitable shape (e.g., a semicircle, an ellipse, an oval, a square, etc.). In this example, the end of the tube **1912** points away from the boost body **1920**. However, in other examples, the tube **1910** may be formable in a

different direction, or, additionally or alternatively, pointed in a predetermined direction other than that depicted in FIGS. **19**A-**19**B. In this example, the tube **1910** emerges from the midpoint between holes **1924** and **1926** on the boost body **1920**. In other examples, the tube **1910** may emerge from another location along the boost body **1920**. While FIGS. **19**A-**19**B depict one tube **1910**, more than one tube may be utilized in other implementations.

(61) FIGS. **20**A-**20**B are perspective views of an example implementation of a unitary assembly of an example tube **2010** and an example boost body **2020**. While not shown in the example of FIGS. **20**A-**20**B, the example tube **2010** continues from the bottom side of the boost body **2020**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **2010** connects into the boost body **178**. In this example, the tube **2010** further includes a first tube **2030** and a second tube **2040** that extend from different positions on the boost body **2020** and converge to form a third tube **2010**. In this example, the first tube **2030** and the second tube **2040** are supported by a first arm **2032** and a second arm **2042**, respectively, the first tube **2030**, the first arm **2032**, and the boost body (i.e., the boost body **2020**) forming a unitary structure and the second tube **2040**, the second arm **2042**, and the boost body **2020** forming a unitary structure. While there are two arms **2032**, **2042** shown in this example, in other implementations there may be one arm or more than two arms. In this example, the arms **2032** and **2042** are web-like structures that provide structural support (e.g., strength and rigidity) to the tube **2010**. However, the arms **2032** and **2042** may have other shapes, geometries, etc. to provide the same or a similar function. In this example, the boost body **2020** has holes **2022**, **2024**, **2026** that may be utilized to fasten the boost body **2020** to the boost body **178**. In other examples, one hole, two holes, or more than three holes may be utilized to fasten the boost body **2020** to the boost body **178**. In some examples, the arms **2032**, **2042** may be manufactured using a separate additive manufacturing process than the process to unify the tube **2010** and the boost body **2020**. The separate additive manufacturing process may be employed to print parts on a previously machined piece. In this example, the boost body **2020** is shaped as a partial circle. However, in other examples, the boost body **2020** may be any other suitable shape (e.g., a semicircle, an ellipse, an oval, a square, etc.). In this example, the end of the tube **2012** points away from the boost body **2020**. However, in other examples, the tube **2010** may be formable in a different direction, or, additionally or alternatively, pointed in a predetermined direction other than that depicted in FIGS. **20**A-**20**B. In this example, the tubes **2030**, **2040** emerge from two locations opposite each other on the boost body **2020**. In other examples, the tubes **2030**, **2040** may emerge from various locations on the boost body **2020** and relative positions to each other. While FIGS. **20**A-**20**B depict two tubes **2030**, **2040** forming into one tube **2010**, more than two tubes may alternatively join to form more than one tube in other implementations.

(62) FIGS. **21**A-**21**B are perspective views of an example implementation of the unitary assembly of the example tube **2110** and the example boost body **2120**. While not shown in the example of FIGS. **21**A-**21**B, the example tube **2110** continues from the bottom side of the boost body **2120**, similar to as shown in FIG. **11**A. Therefore, in this implementation, the tube **2110** connects into the boost body **178**. In this example, the boost body **2120** has four holes **2122**, **2124**, **2126**, **2128** that may be utilized to fasten the boost body **2120** to the boost body **178**. In other examples, one hole, two holes, three holes, or more than four holes may be utilized to fasten the boost body **2120** to the boost body **178**. In this example, the boost body **2120** is shaped as a circle. However, in other examples, the boost body **2120** may be any other suitable shape (e.g., a semicircle, an ellipse, an oval, a square, etc.). In this example, the end of the tube **2112** points away from the boost body **2120**. However, in other examples, the tube **2110** may be formable in a different direction, or, additionally or alternatively, pointed in a predetermined direction other than that depicted in FIGS. **21**A-**21**B. In this example, the tube **2110** emerges from the midpoint between holes **2122** and **2124** on the boost body **2120**. In other examples, the tube **2110** may emerge from another location on the boost body **2120**. While FIGS. **21**A-**21**B depicts one tube **2110**, more than one tube may be utilized in other implementations. Further, while arms, fillets, or other support structures are not pictured in

the example of FIGS. **21**A-**21**B, other implementations of the structure of FIGS. **21**A-**21**B may include support structures of varying shapes and sizes between the tube **2110** and the boost body **2120**.

(63) Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name. As used herein, “approximately” and “about” refer to dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections.

(64) “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

(65) As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

(66) From the foregoing, it will be appreciated that example systems, apparatus, and articles of manufacture have been disclosed that provide a plug assembly with a relatively long pitot tube, which enables the pressure sense chamber to sense pressure further downstream of the valve. The examples disclosed herein also enable the tube to be re-oriented based on the orientation of the valve body and actuator casing.

(67) Examples and example combinations disclosed herein include the following:

(68) Example 1 includes a pressure regulator comprising a valve body defining a fluid passageway

between an inlet and an outlet, a seat in the fluid passageway, an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body, and a plug assembly including a stem, a plug coupled to the stem, a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway, the stem guide having a flange with a notch, a boost body on the second side of the stem guide, and a tube extending through the boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the first end aligned with the notch and the second end disposed in a downstream location relative to the seat such that the pressure sense chamber is in fluid communication with the downstream location, wherein the boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.
(69) Example 2 includes the pressure regulator of example 1, wherein a portion of the tube between the boost body and the second end is curved.
(70) Example 3 includes the pressure regulator of example 2, wherein the portion of the tube between the boost body and the second end is formable.
(71) Example 4 includes the pressure regulator of example 2, wherein the portion of the tube is supported by a first arm of the boost body and a second arm of the boost body.
(72) Example 5 includes the pressure regulator example 4, wherein the first arm and the second arm of the boost body are unitary with the boost body and the tube.
(73) Example 6 includes the pressure regulator of example 2, wherein the portion of the tube is bent to form a first portion and a second portion, the first portion nearest to the boost body and the second portion nearest to the second end of the tube, and an arm connects the first portion and the second portion, the arm, the first portion, and the second portion forming a unitary structure.
(74) Example 7 includes the pressure regulator of example 2, wherein the portion of the tube is bent to form a first portion, a second portion, and a third portion, wherein the first portion is at a first orientation, the second portion is at a second orientation different than the first orientation, and the third portion is at a third orientation different than the first and second orientations.
(75) Example 8 includes the pressure regulator of example 1, wherein the tube further includes a first tube and a second tube that extend from different positions on the boost body and converge to form a third tube.
(76) Example 9 includes the pressure regulator of example 8, wherein the first tube and the second tube are supported by a first arm and a second arm, respectively, the first tube, the first arm, and the boost body forming a unitary structure and the second tube, the second arm, and the boost body forming a unitary structure.
(77) Example 10 includes a pressure regulator comprising a valve body defining a fluid passageway between an inlet and an outlet, a seat in the fluid passageway, an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body, and a plug assembly including a stem, a plug coupled to the stem, a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway, the stem guide having a flange with a notch, and means for orienting a boost body with a tube from a first orientation to a second orientation relative to the valve body.
(78) Example 11 includes the pressure regulator of example 10, wherein a first end of the tube is disposed in the notch.
(79) Example 12 includes the pressure regulator of example 11, wherein the means for rotating the boost body is moveable away from the stem guide to move the first end of the tube out of the notch.

(80) Example 13 includes the pressure regulator of example 10, wherein the means for rotating the boost body includes a flange of the stem guide that has four notches, wherein the boost body can be rotated to four positions to align the first end of the tube with a corresponding one of the notches.
(81) Example 14 includes the pressure regulator of example 10, wherein the tube is curved to include a 90° turn.
(82) Example 15 includes a plug assembly for a pressure regulator, the plug assembly comprising a boost body on a side of a stem guide, and a tube extending through a boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the first end aligned with a first notch on the stem guide and the second end to be disposed in a downstream location relative to a seat in a fluid passageway such that a pressure sense chamber is in fluid communication with the downstream location, wherein the boost body is moveable away from the side of the stem guide and rotatable relative to the stem guide to position the tube in a second notch on the stem guide.
(83) Example 16 includes the plug assembly of example 15, wherein a portion of the tube between the boost body and the second end is curved.
(84) Example 17 includes the pressure regulator of example 16, wherein the portion of the tube is supported by a first arm of the boost body and a second arm of the boost body, the first arm and the second arm are unitary with the boost body and the tube.
(85) Example 18 includes the pressure regulator of example 16, wherein the portion of the tube is curved to form a first portion and a second portion, the first portion nearest to the boost body and the second portion nearest to the second end of the tube, and wherein an arm connects the first portion and the second portion, the arm, the first portion, and the second portion forming a unitary structure.
(86) Example 19 includes the pressure regulator of example 16, wherein the portion of the tube is curved to form a first portion, a second portion, and a third portion, wherein the first portion is at a first orientation, the second portion is at a second orientation, and the third portion is at a third orientation.
(87) Example 20 includes the pressure regulator of example 1, wherein the tube further includes a first tube and a second tube that extend from different positions on the boost body and converge to form a third tube, the first tube and the second tube are supported by a first arm and a second arm, respectively, the first tube, the first arm, and the boost body forming a unitary structure and the second tube, the second arm, and the boost body forming a unitary structure.
(88) Although certain example systems, methods, apparatus, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, methods, apparatus, and articles of manufacture fairly falling within the scope of the claims of this patent.
(89) The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Claims

1. A pressure regulator comprising: a valve body defining a fluid passageway between an inlet and an outlet; a seat in the fluid passageway; an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body; and a plug assembly including: a stem; a plug coupled to the stem; a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway; a boost body on the second side of the stem guide; and a tube extending through the boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube

having a first end and a second end, the second end disposed in a downstream location relative to the seat such that the pressure sense chamber is in fluid communication with the downstream location, wherein a portion of the tube between the boost body and the second end is curved and supported by a first arm and a second arm of the boost body, and wherein the boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.

2. The pressure regulator claim 1, wherein the first arm and the second arm of the boost body are unitary with the boost body and the tube.

3. The pressure regulator of claim 1, wherein the portion is a first portion, and wherein a second portion of the tube is straight.

4. A pressure regulator comprising: a valve body defining a fluid passageway between an inlet and an outlet; a seat in the fluid passageway; an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body; and a plug assembly including: a stem; a plug coupled to the stem; a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway; a boost body on the second side of the stem guide; and a tube extending through the boost body, the tube and the boost body forming a first unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the second end disposed in a downstream location relative to the seat such that the pressure sense chamber is in fluid communication with the downstream location, wherein a portion of the tube between the boost body and the second end is curved, wherein the portion of the tube is curved to form a first portion and a second portion, the first portion nearest to the boost body and the second portion nearest to the second end of the tube, and a web connects the first portion and the second portion, the web, the first portion, and the second portion forming a second unitary structure, and wherein the boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.

5. The pressure regulator of claim 4, wherein the portion of the tube is curved to form the first portion, the second portion, and a third portion, wherein the first portion is at a first orientation, the second portion is at a second orientation different than the first orientation, and the third portion is at a third orientation different than the first and second orientations.

6. A pressure regulator comprising: a valve body defining a fluid passageway between an inlet and an outlet; a seat in the fluid passageway; an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body; and a plug assembly including: a stem; a plug coupled to the stem; a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway; a boost body on the second side of the stem guide; and a tube extending through the boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the second end disposed in a downstream location relative to the seat such that the pressure sense chamber is in fluid communication with the downstream location, wherein the tube includes a first tube and a second tube that extend from different positions on the boost body and converge to form a third tube, and wherein the boost body is rotatable with respect to the stem guide to position the tube in at least one other orientation relative to the actuator casing.

7. The pressure regulator of claim 6, wherein the first tube and the second tube are supported by a first arm and a second arm, respectively, the first tube, the first arm, and the boost body forming a first unitary structure and the second tube, the second arm, and the boost body forming a second unitary structure.

8. A pressure regulator comprising: a valve body defining a fluid passageway between an inlet and an outlet; a seat in the fluid passageway; an actuator casing coupled to the valve body, the actuator casing defining a pressure sense chamber and a channel between the pressure sense chamber and the fluid passageway in the valve body; and a plug assembly including: a stem; a plug coupled to the stem; a stem guide disposed in the channel of the actuator casing, the stem moveable in the stem guide to move the plug relative to the seat, the stem guide having a first side facing the pressure sense chamber and a second side facing the fluid passageway; and a key to orient a boost body with a tube from a first predetermined orientation to a second predetermined orientation relative to the valve body, wherein the key includes at least one projection that corresponds to a notch in the boost body.
9. The pressure regulator of claim 8, wherein a first end of the tube is disposed in the boost body.
10. The pressure regulator of claim 9, wherein the boost body includes at least one notch to align with the tube.
11. The pressure regulator of claim 8, wherein the tube is curved to include a 90° turn.
12. A plug assembly for a pressure regulator, the plug assembly comprising: a boost body on a side of a stem guide; and a tube extending through the boost body, the tube and the boost body forming a unitary structure to prevent movement of the tube relative to the boost body, the tube having a first end and a second end, the second end to be disposed in a downstream location relative to a seat in a fluid passageway such that a pressure sense chamber is in fluid communication with the downstream location, wherein the boost body is moveable away from the side of the stem guide and rotatable relative to the stem guide to position the tube.
13. The plug assembly of claim 12, wherein a portion of the tube between the boost body and the second end is curved.
14. The plug assembly of claim 12, wherein a portion of the tube between the boost body and the second end is straight.
15. The plug assembly of claim 12, wherein the first end of the tube is disposed in the boost body.
16. The plug assembly of claim 15, wherein the boost body includes at least one notch to align with the tube.
17. The plug assembly of claim 12, wherein the boost body is rotated relative to a key including at least one projection that corresponds to a notch in the boost body.